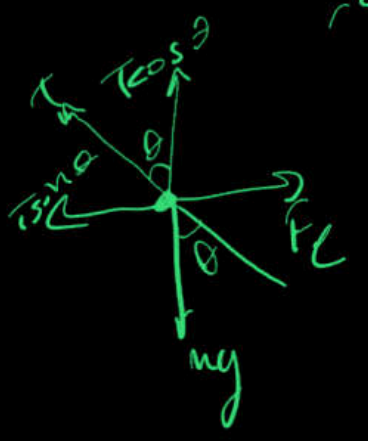
1) $F = k \cdot \frac{|q_1||q_2|}{r^2}$ $\quad F_c = 0,8\,N$



$\frac{F_c}{mg} = \tan\theta = \frac{0,8}{0,6} = 1,33$ $\quad \arctan(1,33) = \theta \cong 53°$

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2) $E = k\frac{q}{r^2} \leftarrow \oint E\,dA$

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3) $V_D = V_1 + V_2 + V_3 = k\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3}\right)$

$U = k\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right) = -3,15\,J$

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4) $\frac{1}{15} + \frac{1}{3} = \frac{1}{C_{eş}} = \frac{6}{15} = \frac{2}{5} \Rightarrow C_{12} = 2,5\,\mu F$

$C_{(12)} = 2,5 + 6 = 8,5\,\mu F$

$\frac{1}{8,5} + \frac{1}{10} = \frac{1}{C_{eş}} \Rightarrow 5,96\,\mu F$

F. U

$q = C \cdot U \rightarrow Q_t = 5,96 \cdot 15 = 89,5\,\mu C$

C_{123} ile C_4 seri bağlı $\Rightarrow$ ikisi de $89,5\,\mu C$

C_4
$20\,\mu F \rightarrow$ (U) $= \frac{q}{C} = \frac{89{,}5\,\mu C}{20} = 4{,}47V$ $\quad 15 - 4{,}47 = 10{,}53V$

C_{123}

C_3 tương tự $\Rightarrow q = 6 \cdot 10{,}53 = 63{,}2\,\mu C$

$q_{12} = q_1 = q_2$ (song song) $= 89{,}5 - 63{,}2$
$= 26{,}3\,\mu C$
